WEEK 4 DAY 24

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Final Technical Review

Problem and data

Problem	DATA
Goal	Estimate state of charge (SOC, %) from live sensor readings
Secondary goal	Estimate remaining charging time (seconds)
Cell / source	LG 18650HG2 Li-ion, 25 °C folder, 36 test files
Rows	104,560 (every 10th sample kept)
Target	SOC = 100 × (1 + capacity_Ah / 3.0), clipped to 0–100
Inputs (5)	voltage_v, current_a, cell_temp_c, power_w, ambient_temp_c
Excluded (leakage)	capacity_ah, wh_accu, cnt, step, cycle, cycle_level, procedure, status, prog_time, step_time
Metrics	MAE, RMSE, R^2 (SOC %)

MODEL COMPARISON

METHOD	FEATURES	MAE	RSME	R^2
Random Forest	Raw	3.284	10.474	0.848
XG Boost	Raw	3.301	10.082	0.859
Linear	Raw	3.473	4.586	0.971
XG Boost	Engineered	22.599	27.527	-0.049
Random Forest	Engineered	22.943	28.917	-0.158
Mean Baseline	---	23.041	27.509	-0.048
Linear	Engineered	23.193	27.771	-0.068
MLP(ANN)	Engineered	25.932	31.611	-0.384

Overfit check

Model	Train MAE	Test MAE	Gap	Train R^2	Test R^2	Verdict
RandomForest — raw	0.884	3.284	2.400	0.995	0.848	possible overfitting
XGBoost — raw	1.325	3.301	1.975	0.992	0.859	possible overfitting
RandomForest — engineered	19.101	22.943	3.842	0.290	-0.158	underfitting (both scores weak)
XGBoost — engineered	20.949	22.599	1.649	0.202	-0.049	underfitting (both scores weak)

Random forest validation

depth	min leaf	trees	validation MAE
8	4	300	1.2431
8	4	450	1.2443
8	1	450	1.2474
12	4	450	1.4586
12	1	300	1.5772
16	1	300	1.6694

Tuned vs untuned Random forest

Test result	MAE	RMSE	R^2
Untuned RF (Day 16)	3.284	10.474	0.848
Tuned RF (Day 19/20)	2.899	8.379	0.903
Improvement	−11.7 %	—	—

Best case:

Depth 8,min leaf 4,300 trees.

FINAL MODEL

Model	CV MAE	CV RMSE	CV R^2	Worst fold MAE
Mean baseline	24.593	28.304	-0.122	32.328
Linear (raw)	3.176	5.091	0.962	3.526
RandomForest (tuned)	**2.126**	**4.763**	**0.965**	**2.668**
HistGradBoost (raw)	1.998	5.088	0.959	2.639

Error Analysis

- Error_by_soc_band.csv

SOC band	Rows	MAE	Max error
0-15 (knee)	**2814**	**10.514**	**88.5**
15-30	4268	1.829	52.7
30-50	4075	3.147	34.1
50-70	7742	1.635	29.6
70-85	4418	1.959	14.3
85-100	4451	2.016	84.1

Finding	Value
Worst SOC band	0–15 % (knee): MAE 10.51, 5× the other bands
Worst files	25degC_Charge6, 25degC_HPPC (7.9 over 4,805 rows)
Error concentration	Top 1 % of rows carry 75 % of the squared error
RMSE / MAE	2.89 → rare but large mistakes

Remaining charging time, and limitations

- Test: 2 charge files, 482 rows (Day 23 output)

Charging time, test	MAE (s)	RMSE (s)	R^2
RF frozen (100 trees, depth 10)	24,390	33,440	−10,850
Mean baseline	5,759	5,768	−322

The model is worse than guessing the average → charging time is reported as NOT validated

Reproducible input to output workflow

Step	Stage	Detail
1	Raw sensor CSV	voltage_v, current_a, cell_temp_c, power_w, ambient_temp_c
2	Select 5 raw features	ColumnTransformer by name; no scaling
3	Frozen pipeline	soc_final_pipeline.joblib (RandomForest, 300 trees, depth 8, min leaf 4, seed 42)
4	Clip	np.clip(prediction, 0, 100)
5	Output	SOC (%) per row

THANK YOU